Image Registration is the process of making an image conforms to another image. A map coordinate system is not necessarily involved in image registration. For example, if image A is not rectified and it is being used with image B, then image B must be registered to image A so that they conform to each other. In this example, image A is not rectified to a particular map projection, so there is no need to rectify image B to a map projection.

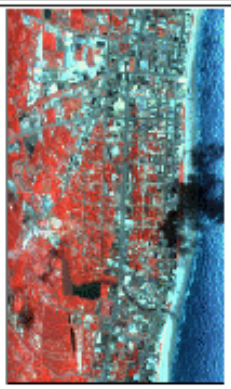
Georeferencing refers to the process of assigning map coordinates to image data. The image data may already be projected onto the desired plane, but not yet referenced to the proper coordinate system. Rectification, by definition, involves georeferencing, since all map projection systems are associated with map coordinates. Image-to image registration involves georeferencing only if the reference image is already georeferenced. Image to image registration does not involve georeferencing if reference image does not have standard map projection (georeferenced). Georeferencing, by itself, involves changing only the map coordinate information in the image file. The grid of the image does not change.

Objectives

- To rectify a raw digital image using
  a) *image-to-map* rectification;
  b) *image-to-image* registration; and/or
  c) *Rectification* using GPS-derived ground control.

Remotely sensed images store important geographic information. Unfortunately, unless they are rectified, the spatial information within the image can only be referenced using simple row and column coordinate information rather than true geographic coordinates (e.g., in latitude/longitude) or coordinates in a map projection (e.g., Universal Transverse Mercator). Geometric rectification (georeferencing) is the process whereby the geometry of an image is made *planimetric*, allowing the data to be used to extract accurate map location information for features in the data. The process usually involves identifying the x, y coordinates of real-world **ground control points** (GCPs) such as road intersections, the intersection of fence lines, etc., and the row and column coordinates of these same features within the image. Geometric rectification is necessary whenever accurate area, direction, and distance measurements are to be extracted from the rectified imagery. Most serious earth science remote sensing research is based on analysis of data that have been accurately rectified to a base map.

Method I – Image-to-Map Rectification

File Name	*myrtle-beach_cams_1997-08-02.img*	Quick View
Location	Myrtle Beach, SC	 Altitude 1,524 m AGL
Sensor	Calibrated Airborne Multispectral Scanner (CAMS)	
Spatial	3 ?3 m	
Temporal	August 2, 1997	
Spectral	Layer 1: Band 2 = Green (.52-.60) Layer 2: Band 4 = Red (.63-.69) Layer 3: Band 6 = NIR (.76-.90)	

Two basic operations must be performed in order to geometrically rectify a remotely sensed image to a map coordinate system:

1. Spatial Interpolation, which defines the nature of the mathematical geometric coordinate transformation that must be applied to fill every pixel in the rectified output matrix with values from the unrectified original input matrix.

2. Intensity Interpolation, which is the method used to determine the brightness value to be assigned to each pixel in the rectified output image.

ERDAS Imagine offers several methods to rectify images to maps. In this exercise you will perform a simple rectification of a Calibrated Airborne Multispectral Scanner (CAMS) image of downtown Myrtle Beach, SC to a UTM map projection using nearest-neighbor interpolation. The procedure follows the general outline:

1. The **GCP Editor** is used to create ***.gcc** files containing image (row and column values) that relate to map coordinates (UTM *x* and *y* coordinates in meters) for selected ground control points.

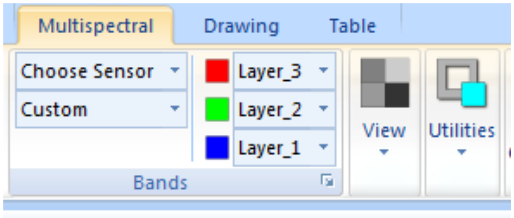
2. The ***.gcc** file is input to the **Transformation Editor**, which creates a small matrix containing the transformation coefficients.

3. A geometric model file **(*.gms)** is then created using the model properties and the original ***.img** image file is input to the **Resample Display** program to produce a rectified ***.img** image file.

Subset the Image

Calibrated Airborne Multispectral Scanner (CAMS) data are well-known for geometric irregularities due to the side-to-side scanning motion of the multispectral scanner mirror and the unstable airborne (suborbital) platform. In this exercise, you will minimize these errors by subsetting and then rectifying a small area of the Myrtle Beach image. Place the ***myrtle-beach_cams_1997-08-02.img*** file in the viewer with RGB = bands 3,2,1.

Now click on the |**Multispectral | Utilities| Subset & Chip|** icon in the Imagine icon panel.



This brings up the **GLT Subset** dialog box. Specify **myrtle-beach_cams_1997-08-02.img** as the input file and name the output file **myrtle-beach_cams_1997-08-02_subset.img**.

Specify **Type = File** as you are subsetting an unrectified image that does not have map coordinates. Now you are going to subset the original image using specific file coordinates. Specify in the **Subset Definition:**

ULX 38 **ULY** 5
LRX 450 **LRY** 400

There are several parameters you can adjust under Options, but just use **Output: unsigned 8-bit** and select all three bands, i.e., **Select Layers: 1:3.** Once you have entered the above coordinates, click **Apply**. This creates the subset imaging consisting of three bands. Display the subset image to make sure the subset was successful.

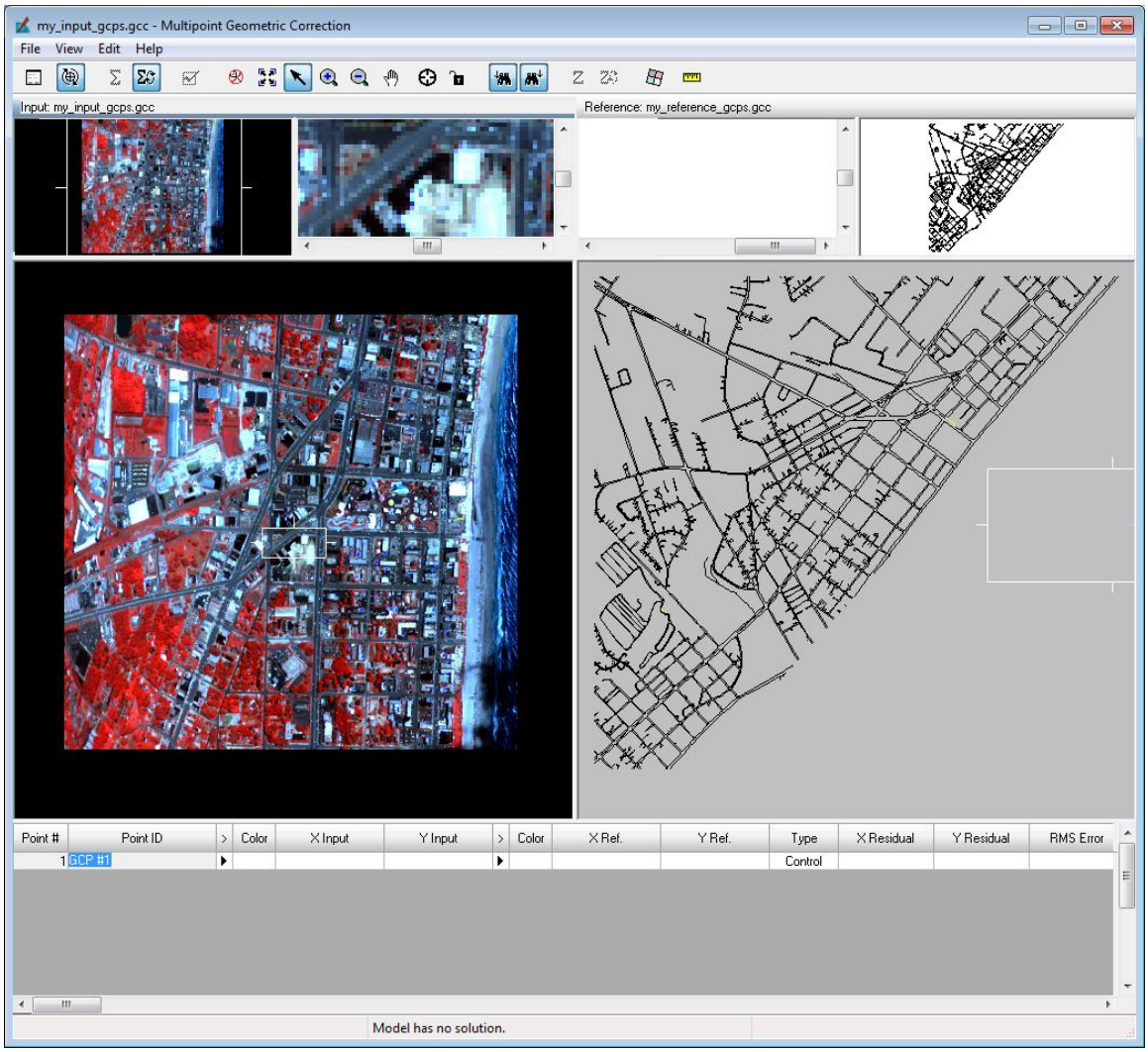
Becoming Familiar displaying Image and Vector Data

You are now going to use a U.S. Geological Survey Digital Line Graph (DLG) of the road network in Myrtle Beach to rectify your image.

Open the subset Myrtle Beach image file in **2D View #1**. Open a **new 2D View** (top left icon). Now open a **Vector layer** in the **2D View #2**. When the menu appears, set the **Files of type to Arc Coverage**. This will display the **mb_bldgs** and the **mb_roads** vector files. Select **mb_roads**. It may be difficult to see it, so right click in the viewer and **zoom to data extent**. Also, right click in the viewer and **select background color** and set it to gray. The road network should now be visible on the black background. This was just practice to introduce you to having a couple of windows open with a) an unrectified image, and b) some vector information.

GCP Selection

Now, display the Myrtle Beach subset image in a new viewer. Go to |**Multispectral** | **Transform & Orthocorrect | Control Points** |. When the **Set Geometric Model** menu appears, specify **Model List: Polynomial**. This automatically launches the **"Multipoint Geometric Correction"** module. When the **GCP Tool Reference Setup** appears, select **Vector Layer** (new Viewer) and press OK. When the **Reference Vector Layer** menu appears, select the **mb_roads** file and click OK. The **Reference Map Information** appears, click OK. Your viewer should look something like the screenshot below.



When the **Polynomial Model Properties** (No File) menu appears, make sure the polynomial order =1 and save it as "yourmodel.gms". Then close it. You may have to right-click in the image input viewer area to set the **band combinations** correctly (e.g., bands = 3,2,1).

Before you go any further, go to "File" and Save Input As" and give it some name like "My_input_file_GCPs.gcc". Then go to "File" and "Save Reference As" and give it some name

like "My _reference_file_GCPs.gcc". You get a warning message, but don't worry about it. Save these two files quite often as you collect ground control points.

Notice that when you move your cursor inside the road network that the UTM projection coordinates are in meters northing and easting. When you move the cursor into the unrectified image area, the coordinates are in simple row and column coordinates.

You will now select Ground Control Points (GCPs) using UTM coordinates for image rectification. GCP's consist of two pairs of x,y coordinates:

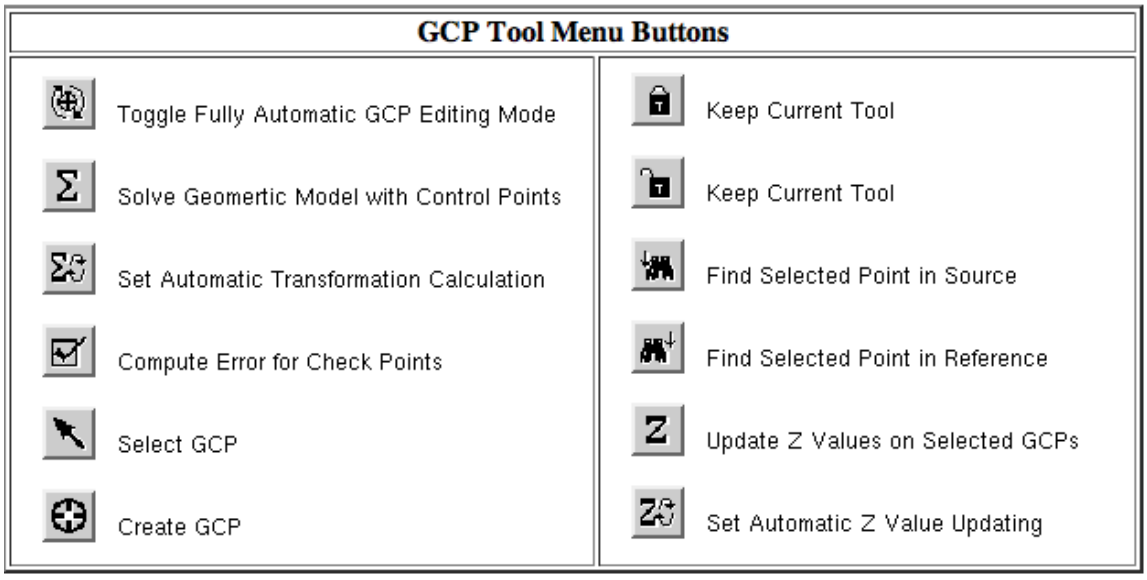
- **Source or input coordinates**, which are data file coordinates associated with GCPs located in the image to be rectified, and

- **Reference or output coordinates**, associated with the planimetric map or already rectified reference image to which the source image is being rectified.

When selecting GCP's, collect points evenly distributed throughout the **entire** area to be georeferenced. Don't collect points in just a small region. This will usually yield a good rectification. Features like road intersections, corners of large building complexes, and land/water points etc. are good choices for finding locations on both images and topographic maps. A good number of GCP's for this exercise is 10-15. In a real research project the number should normally be >50. Now go to the cross-hair cursor and click it just once:



The crosshair is for locating a GCP in either the unrectified image or the reference viewer. [If you had no reference map or image, you would have to enter the coordinates by hand or enter a GCP file collected with a Global Positioning System (GPS)]. Locate GCP #1 by clicking on the crosshair button and then moving to the CAMS image and clicking on the location (street intersection) you want to place the GCP. Now click on the crosshair button again, but this time locates a GCP at the corresponding point on the DLG road coverage. [You should set the background color of the road network to gray or black by placing the cursor in the viewer and right clicking to get the **Quick View** menu and then click on **Background Color**.] Remember that the road coverage represents street centerlines so placement of GCPs should take this into consideration. When you are finished with the placement of both the source and destination locations for GCP #1, you should be sent to the GCP#2 row automatically. You can fine tune the location of your GCPs by selecting them with your cursor and using the arrow keys to move them to their desired position. Also, you may want to change the color of your GCPs by going to the **GCP Tool** window and placing the cursor over the *Color* row of the corresponding GCP. Hold down the right mouse button over the color and select a new color.

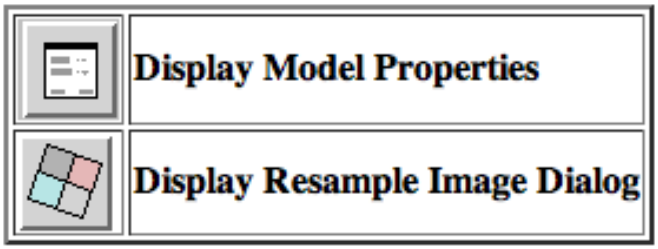


During the GCP selection process, you may wish to use the **Automatic Transformation Calculation** button (see menu above). This will display the coefficients of the transformation matrix and calculate the RMS error between the file points and map points. For this exercise you will try to get the Root Mean Square (RMS) error to be < 1.0. In the GCP Tool window, use the scroll bar at the bottom to move your view to the far right-hand side. Here are the listings of the errors (residuals, RMS error and contribution for each GCP). This can help you determine the GCP's you might wish to change. If the RMS Error for all your GCPs is <1, then congratulations, you are talented at selecting GCP's. The Total RMS Error for all GCPs is listed just right of the buttons in the GCP Tool (for example: (Total) 0.3984)). Try to get the total as close to zero as possible. If the Total RMS Error is >1.0, you will need to either delete a GCP to improve the RMS error or move the GCPs until the RMS error figure is more appropriate. You might want to see the RMS error be automatically calculated by grabbing one of the GCPs with your cursor and dragging it around the image, but be sure to put it back in its proper place!

The procedure for deleting a GCP from the table is to move the cursor to the far left-hand side of the table and in the first box of the row relating to that GCP you select the left mouse button (lmb) (the whole row should turn yellow) then select the rmb and in the Quickview menu that appears and select the **Delete Selection** option. Try not to delete too many GCP's - it's better to try and make them fit before removing them.

Resampling and Rectification

Once your RMS error is approximately < 1.0, you are ready to resample the image. Click on the **Display Model Properties** button in the **main menu (it is the icon on the far left)**. Browse through the Parameters, Transformation, and Projection folders to view the various settings. Now save this geometric model by clicking on **Save as** and naming the file something logical. Now close the **Model Properties** window and click on the **Display Resample Image Dialog** button in the **Geo Correction Tools** window.



When the **Resample** window opens, find **the Output Cell Sizes** boxes. Change both the X and Y to 3.0. (This indicates the size of the pixels for the rectified image; 3.0 x 3.0 m for CAMS Data). Select the nearest neighbor resampling algorithm (it should already be selected). Now name the output image ***ex4.img***, click **OK**, and let the resample model perform. Your rectified image will be created and stored under the filename which you designated. Close down the GCP Tool window by going under **File - Close** and saving all files.

Now start a new viewer to display your rectified image and compare it with the unrectified image. Open up the roads vector layer by right-clicking in the rectified image window, Open Vector Layer, and specify mb_roads or mb_bldgs (remember they are coverage files).

Questions:

1) Describe the difference between the two image rectification techniques you used and the associated errors that might be introduced when using each of these techniques. Give some examples of when you would want to utilize one geometric rectification technique versus the other.

2) Is the spatial distribution of your GCPs (in reference to scene rectification) important to ensure an acceptable rectification? Why or why not?

3) What is your definition of the ideal Ground Control Point (GCP)?